SafeExec

Draft Report, Deck, and Final Test Evidence

A hardened, threat-modeled Python execution sandbox for LLM-agent tool use.

4/4
Fresh W12 checks
smoke, tests, validation, audit

50/50
Docker target-host probes
W8 hardened Docker

49/50
gVisor target-host probes
one timing miss

Hard Stop 6

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Prepared: 2026-06-26

Message

The artifact is technically alive and reproducible locally; the remaining work is corpus depth and final benchmark refresh.

The problem is where generated text becomes process execution.

Agent risk

Prompt injection, excessive agency, and ordinary model errors can become filesystem, network, or resource actions when code is executed.

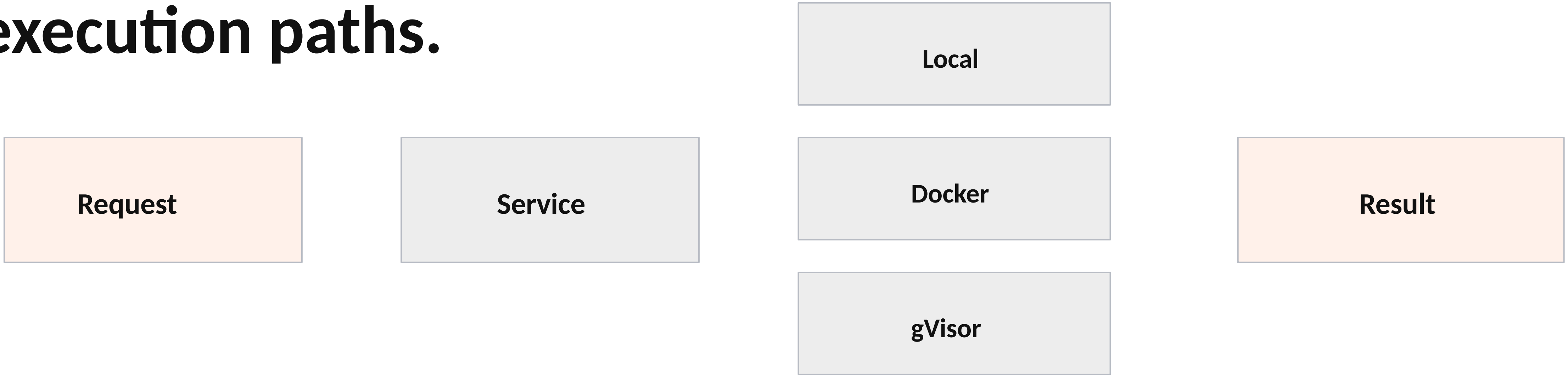
Project response

Keep the artifact narrow: Python-only execution, visible limits, explicit backends, repeatable tests, and honest containment evidence.

Core claim

SafeExec does not assert sandbox safety. It measures the execution boundary and reports what the evidence actually supports.

Architecture: one request model, three execution paths.



Local is a development smoke path. Docker and gVisor are the containment-relevant paths.

Implementation surface is compact and traceable.

models.py	request, limits, result schema
service.py	backend selection and execution entry
backends/docker.py	hardened Docker and gVisor command builder
api/server.py	JSON /health and /execute API shell
run_* scripts	validation, midpoint evidence, final evidence
docs/reproducibility	runbook, environment, data, manifest

The small surface helps the final report trace evidence back to source files instead of hiding behavior in framework layers.

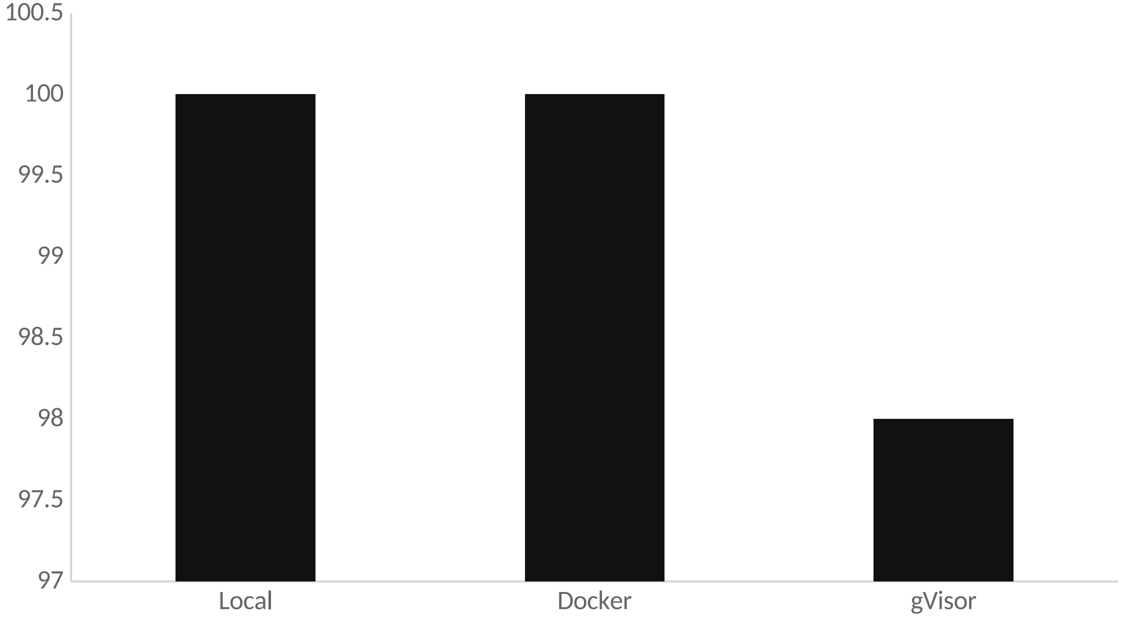
Fresh W12 local evidence passed.

PASS	PASS	12/12	PASS
Smoke	**Unit tests**	**Validation**	**Repro audit**
structured execution result	functional test suite	local cases plus API trace	no blocking findings

Generated by `make final-evidence` into docs/12-hard-stop-6/evidence/.

This is the current local evidence. Docker/gVisor evidence remains target-host evidence because local macOS is not the official container benchmark environment.

Target-host benchmark evidence: Docker stable, gVisor mostly passing.



W8 target-host result

Local: 30/30
Docker: 50/50
gVisor: 49/50

Interpretation: gVisor is integrated, but final timing needs cold-start and warm-run separation.

Final strength depends on evidence growth.

Known gaps

- HumanEval/MBPP subset manifest still needs finalization.
- Adversarial suite still needs approved count/category coverage.
- gVisor timing needs cold-start vs warm-run separation.

Final-week response

- Re-run Docker/gVisor after corpus expansion.
- Update report/deck tables from final outputs.
- Rebuild source package and record final walkthrough.

Current conclusion: technically on track, with final success dependent on corpus and benchmark completion.

What I will submit for this checkpoint.

- Draft-Report-Deck-Final-Test-Evidence.pdf / .docx
- SafeExec-Near-Final-Technical-Report-Draft.pdf / .docx
- SafeExec-Draft-Report-and-Demo-Deck.pptx
- final-test-evidence-appendix.md and evidence/
- walkthrough-script.md, risk log, AI-use log

Closing message The written report, deck, and test evidence now tell the same story.